

VINAY DEVABHAKTUNI

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PROFESSIONAL SUMMARY

Full-stack software engineer with 2+ years of industry experience developing, deploying, testing, and debugging production applications and REST APIs. Proficient in Java, JavaScript, and PHP with strong expertise in database design, software documentation, and collaborative development workflows. Experienced in all phases of the software development lifecycle (requirements gathering, design, implementation, testing, deployment) with demonstrated ability to work independently and with senior team members to deliver business-critical solutions. Recently completed MS in Computer Science with focus on software systems, test automation, and cloud infrastructure.

TECHNICAL SKILLS

- **Programming Languages:** Java, JavaScript, PHP, Python, SQL, Hadoop, HDFS, Hive, Spark ML, PySpark, Spark
- **Web Development:** Spring Boot, Restful API, Flask, Node.js, Angular, React.js
- **Databases and Data Structures:** MySQL, Spark SQL, PL/SQL, NoSQL, PostgreSQL, Database Design, Schema Enforcement, Query Optimization
- **Version Control and CI/CD:** Git, GitHub, GitHub Actions, CI/CD pipelines, Docker, Jenkins, Postman
- **Cloud (AWS):** S3, Glue, Glue Data Catalog, Athena, Lambda, EC2, CloudWatch, IAM, SageMaker, Redshift
- **Testing and Quality Assurance:** Junit, Angular test framework, Test Automation, Data Contract testing
- **Development Tools:** Jira, IntelliJ, Visual Studio
- **Soft skills:** Debugging, Code Documentation, Problem Solving, Critical Thinking, Collaborative Development, Troubleshooting
- **AI Integration:** LLM Evaluation, RAG, FAISS, SentenceTransformers, Claude, Gemini, OpenGPT, Ollama, MCP server

PROFESSIONAL EXPERIENCE

University of North Carolina at Charlotte

Graduate Research Assistant

Charlotte, United States of America · Aug 2024 - Present

- Architected and deployed serverless full-stack application on AWS (S3 → Lambda → Glue Crawler → Athena → Flask/EC2), building end-to-end ingestion-to-presentation pipeline and designing databases with optimized partitioning strategies to support five self-service analytics queries for stakeholders.
- Designed and implemented modular four-stage PySpark/Spark SQL transformation layer with reusable feature engineering modules, tested data contracts to ensure schema correctness, and processed multi-source structured and semi-structured data into curated analytical tables for downstream consumption.
- Deployed a Flask service on EC2 to expose Athena query results (over S3-partitioned data) as a stakeholder-facing analytics dashboard, including front-end UI design and full-stack implementation, completing the serverless ingestion-to-presentation loop
- Automated multi-stage research workflows and data pipelines in Apache Airflow under least-privilege IAM roles with CloudWatch monitoring, eliminating ~15 hours/week of manual operations and improving reliability.
- Co-author an IEEE article on trust engineering for LLM-generated code; define a five-class hallucination taxonomy (H1–H5) with programmatic grounding tests across GPT-4, Claude, and Codex to validate correctness and support production ML model evaluation in CI/CD pipelines.

Cognizant Technology Solutions

Programmer Analyst

Chennai, India · Jun 2023 - Jun 2024

- Engineered full-stack REST APIs in Java Spring Boot microservices architecture (backed by MySQL and NoSQL stores) for business-critical applications; optimized API response latency and throughput through efficient query design and caching strategies.
- Developed and tested backend REST API endpoints using JUnit unit tests and integration testing; collaborated with frontend developers (Angular) to design RESTful interfaces and ensure seamless end-to-end functionality
- Orchestrated end-to-end ETL workflows in Apache Airflow with 20% improvement in scheduling efficiency; deployed via GitHub CI/CD pipelines with automated rollback safeguards, reducing deployment errors by 15%.
- Designed and optimized complex SQL queries and database schemas (MySQL, PostgreSQL); improved query execution performance by 25% using stored procedures, triggers, views, and partition-aware indexing.
- Resolved critical production software and data integration bugs in collaboration with senior engineers, improving pipeline reliability, observability, and system resilience.

Programmer Analyst Trainee

Hyderabad, India · Aug 2022 - Jun 2023

- Engineered Python data ingestion pipelines with 10+ automated validation workflows (schema checks, null-rate thresholds, referential integrity tests); reduced manual reconciliation effort by 40% while maintaining 99.9% data integrity.
- Wrote and executed test automation and quality-assurance workflows, including unit-testing frameworks and Angular test cases for frontend applications, which increased test coverage and caught defects earlier, reducing regression bugs in production
- Collaborated with business analysts, QA teams, and senior developers using JIRA and Confluence to clarify requirements, troubleshoot application issues, and document solutions, enabling faster issue resolution and improving stakeholder satisfaction

EDUCATION

University of North Carolina at Charlotte

Master of Science, Computer Science

- **GPA:** 3.83/4.0

Aug 2024 - Present

Charlotte, NC

CMR Technical Campus

Bachelor of Technology, Computer Science Engineering

- **GPA:** 7.24/10.0

Aug 2018 - Jul 2022

Hyderabad, India

PROJECTS

Manhua Matchmaker manhua-matchmaker.streamlit.app

Jan 2026 - Mar 2026

- Ingested ~10,000 manga records via MangaDex REST and AniList GraphQL APIs using Python and SQL scripts, applying fuzzy matching to deduplicate entries (600× speedup: 10 min → <1.5 s)
- Engineered full-stack recommendation engine with automated daily ETL pipelines, vector-based similarity search (FAISS + MiniLM), and responsive gapless reader UI.
- Implemented data pipeline orchestration, vector indexing, and API endpoints for recommendation serving.
- Utilized GitHub Actions for CI/CD, automated testing, and deployment. Technologies: Python, FAISS, Flask APIs, JavaScript, SQL, GitHub Actions

Job Search Automation github.com/VinayD2028/job-search-automation

Jan 2026 - Apr 2026

- Architected end-to-end job search automation platform with daily scraping (LinkedIn, Indeed), AI-powered resume tailoring, and cover letter generation via local LLM (Ollama/llama3.2).
- Developed Streamlit dashboard for full application lifecycle tracking with Gmail API integration for automated interview invite detection and offer tracking.
- Implemented GitHub Actions for CI/CD pipeline, automated testing, and scheduled data ingestion. Technologies: Python, Streamlit, REST APIs, GitHub Actions, SQLite, Gmail API

AWS Serverless Data Pipeline <https://github.com/VinayD2028/aws-serverless-data-pipeline.git>

Jan 2026 - Apr 2026

- Designed and deployed production-grade serverless ETL pipeline on AWS (S3, Lambda, Glue, Athena) with automated data partitioning and query optimization.
- Built Flask microservice to expose analytics data with dashboard UI; implemented comprehensive error handling and CloudWatch monitoring. Technologies: Python, AWS (S3, Lambda, Glue, Athena), Flask, SQL

HUD FMR Fair Pay Dashboard github.com/VinayD2028/HUD-FMR-Dashboard

Feb 2026 - Apr 2026

- Co-developed a Streamlit analytics application on HUD Fair Market Rents 2025, using SQL for data transformations and Git for collaborative version control

CERTIFICATES

- AWS Academy Graduate - Data Engineering: Amazon Web Services Training & Certification: Completion Dec 2025
- PHP and MySQL: Spoken Tutorial / IIT Bombay
- Introduction to Programming Using JavaScript: Udemy
- Generative AI: Enhance Your Data Analytics Career: IBM
- Python Data Analytics: Meta
- Introduction to Data Analytics: Meta
- A/B Testing in Python: DataCamp